AAKASH UDAI

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Professional Summary

AI Engineer (MCA 2025) with strong foundations in Machine Learning, Deep Learning, and Generative AI. Experienced in building RAG-based LLM systems with hybrid search and re-ranking, real-time computer vision applications, and CNN models. Proficient in Python, TensorFlow, Scikit-learn, FastAPI, and vector databases, focused on developing scalable AI-driven solutions.

Technical Skills

- **Languages**: Python
- **Machine Learning & Deep Learning:** CNNs, Transfer Learning, NLP, Regression, Classification, Feature Engineering, EDA, Fine-tuning (LoRA/QLoRA), RAG, Vector stores (Chroma), Hugging Face Hub
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, OpenCV, NLTK, langchain

Projects

Virtual Dress Room Computer Vision — Real-Time Try-On System

- Built a pose-detection based virtual try-on system achieving 24–30 FPS and ~90% garment alignment.
- *Tech Stack: OpenCV, MediaPipe, Python, Pose Detection, Django*

Teacher Lesson Plan Creator (RAG-Based AI System)

- *Developed a RAG-based lesson plan generator using hybrid search (keyword + semantic retrieval) with cross-encoder re-ranking to improve contextual relevance.*
- *Built scalable FastAPI APIs integrating LangChain, ChromaDB, and Hugging Face LLMs for context-aware content generation.*
- *Tech Stack*: *Django, Python, HTML, CSS, JavaScript, Numpy, Pandas, Scikit Learn*

Alzheimer's Prediction (ML)

- Created a CNN for MRI-based Alzheimer's classification achieving 92% validation accuracy; reduced overfitting by 14% with data augmentation.
- *Tech Stack: TensorFlow, NumPy, Pandas, Matplotlib, Scikit-learn*

Sentiment Analysis (NLP Mini-Project)

- Fine-tuned DistilBERT for Twitter sentiment classification achieving 87% F1 score with LoRA optimization
- *Tech Stack*: *Hugging Face Transformers, Python, Pandas, NLTK*

Education

Master of Computer Application –- 2023 –2025

MohandasCollege of Engineering and Technology, Trivandrum (APJAKTU)

Bachelor of Science in Physics- (Sep 2018 – Apr 2021)
Catholicate College, Pathanamthitta (MG University)

Certifications

Diploma in Data Science and AI – Keltron Knowledge Centre